

App Development Project Report

App Dev Project Report

1. Student Details

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About Me: I am a dual degree student currently pursuing B.E. in CSE(AIML) at R V College Of Engineering, Bangalore. My interests include backend development, AIML applications and competitive programming.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web-based Placement Portal Application that allows institutes, companies, and students to efficiently manage campus recruitment activities such as placement drives, job applications, company approvals, student approvals, etc.

Approach:

The app was built using Flask as the backend framework with a secure, role-based structure. It allows the institute admin to oversee recruitment, companies to post placement drives and manage applicants, and students to easily apply and track their placement status.

3. AI/LLM Declaration

I used **Gemini 3 Pro** to assist in checking the code for any errors or bugs, verifying if all the requirements were met, verifying and fine tuning the DB schema and the backend logic .

The extent of AI/LLM usage is around **20%**, limited to **suggestions and code review**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Werkzeug	Securely hash and verify user passwords
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

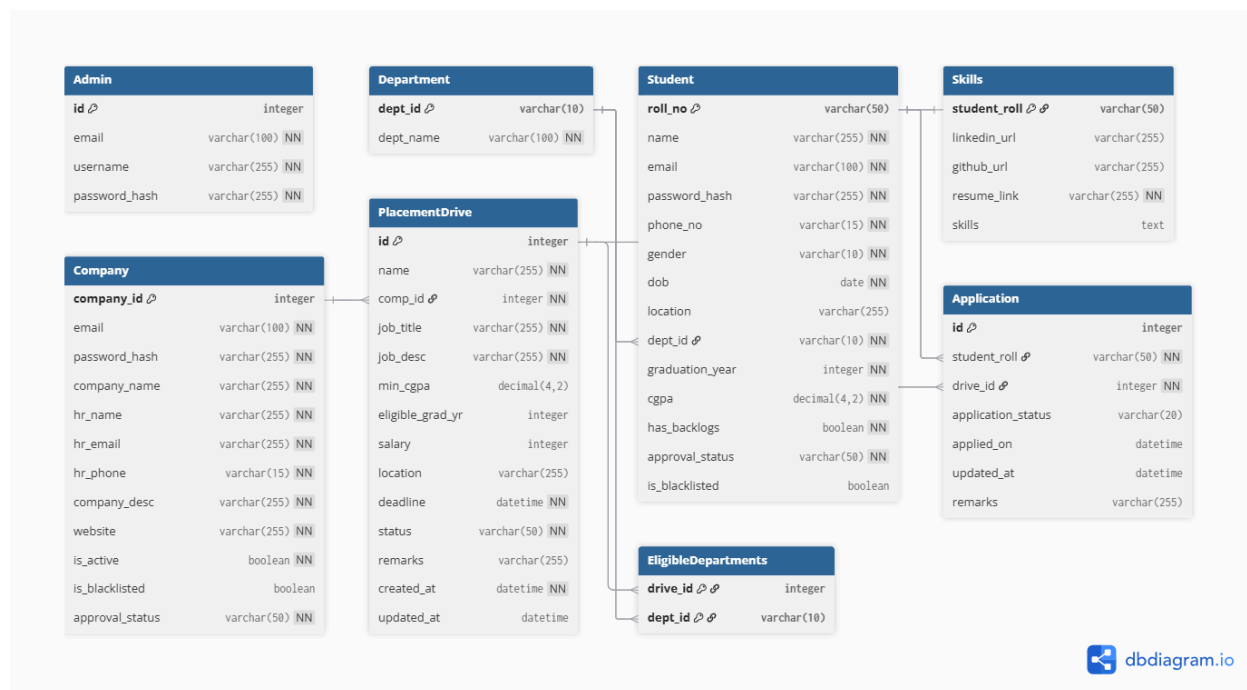
Tables:

1. **Admin** - Stores institute superuser credentials and details (id, email, username, password_hash)
2. **Department** - Stores the departments in the institute (dept_id, dept_name)
3. **Student** - Stores student profile, academic records, and account status (roll_no, name, email, password_hash, phone_no, gender, dob, location, dept_id, graduation_year, cgpa, has_backlogs, approval_status, is_blacklisted)
4. **Skills** - Stores student professional portfolios, technical skills, and resume links (student_roll, linkedin_url, github_url, resume_link, skills)
5. **Company** - Stores registered company profiles, HR contact details, and approval status (company_id, email, password_hash, company_name, hr_name, hr_email, hr_phone, company_desc, website, is_active, is_blacklisted, approval_status)
6. **PlacementDrive** - Stores job postings, eligibility criteria, and drive statuses created by companies (id, name, comp_id, job_title, job_desc, min_cgpa, eligible_grad_yr, salary, location, deadline, status, remarks, created_at, updated_at)
7. **EligibleDepartments** - Maps which academic departments are eligible for specific placement drives (drive_id, dept_id)
8. **Application** - Logs student job applications and tracks their hiring progress (id, student_roll, drive_id, application_status, applied_on, updated_at, remarks)

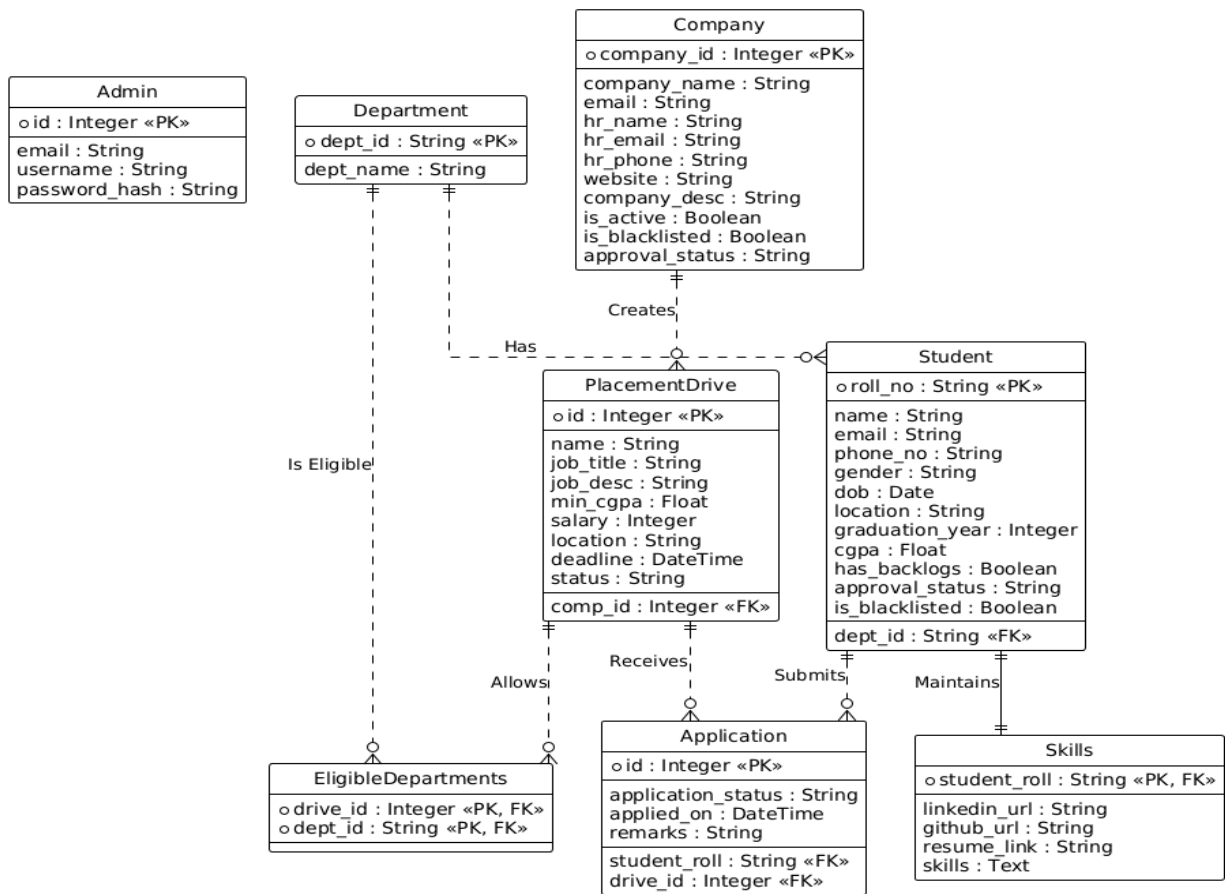
Relationships:

- **One-to-Many** → Department → Student
- **One-to-One** → Student → Skills
- **One-to-Many** → Company → PlacementDrive
- **One-to-Many** → Student → Application
- **One-to-Many** → PlacementDrive → Application
- **One-to-Many** → PlacementDrive → EligibleDepartments
- **One-to-Many** → Department → EligibleDepartments

Database Schema :



ER Diagram :



6. API Resource Endpoints

Endpoint	Method	Description
/login	POST	Authenticate user and generate session
/student/register	POST	Register a new student profile
/company/register	POST	Register a new company profile
/company/drive/create	POST	Create a new placement drive
/student/drive/<id>/apply	POST	Submit the student's application to a drive

<code>/company/application/<id>/review</code>	POST	Update the student's application status (Shortlisted, Hired, Rejected)
<code>/admin/applications</code>	GET	Fetch the master log of all student applications
<code>/student/company/<id></code>	GET	Fetch a company's job profile and list all of their active drives

7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – Main Flask application entry point containing all routing, controllers and backend logic
- **/models** – database models using SQLAlchemy
- **/templates** – Jinja2 HTML templates

Implemented Features:

- Secure user registration, authentication, and role-based access control (Admin, Company, Student).
 - Admin dashboard with gatekeeping workflows for approving or rejecting company registrations.
 - Company portal for creating placement drives, setting eligibility criteria, and reviewing applications.
 - Student portal for searching approved placement drives, submitting applications, and uploading resumes.
 - Application status tracking (Applied / Shortlisted / Hired / Rejected).
 - Duplicate application prevention and database validation constraints.
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8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1bdDpyGL8N1ONjqCA6JWY7yIRj7eNY42/view?usp=sharing>
